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SUMMARY

AI/ML engineer and researcher specializing in security systems, DL infrastructure, and optimizer theory. Solo author of an ICLR 2027 paper on spectral optimizer geometry (OpenReview #8176). Founder of EdgeCI (YC W27 applicant), an Apple Silicon inference regression gate adopted upstream by RapidMLX. 1st-place solo winner of the Trae AI Hackathon, global finalist in the Meta PyTorch OpenEnv Hackathon (solo, 70k+ teams), QuantCraft finalist, and Smart India Hackathon semi-finalist. Builds in C++, Go, Python, Rust, Swift, and TypeScript.

RESEARCH & PUBLICATIONS

- When Does Adaptive Spectral Normalization Define a Fixed Convex Geometry? / Solo Author** ICLR 2027 — Under Review
- Determined when spectral optimizers (**Muon**, **Newton-Schulz**, **Jordan polynomials**) admit a fixed convex potential. Proved one cubic Newton-Schulz step is norm-derived but repeated steps and all positive-depth Taylor-quintic compositions are not, despite preserving order.
 - Derived a sharp filter-preserving condition-number frontier. Verified all critical points via exact **FLINT/Arb** rational certificates over $\mathbb{Q}$ and dependent replay of 4,728 stored spectra. Registered abstract **#8176** on OpenReview.

FOUNDER EXPERIENCE

- EdgeCI — Founder & Core Engineer (YC W27 Applicant)** 2026 – Present
- Built a statistical performance-release gate for **llama.cpp** (121k+ ★) and MLX on Apple Silicon. Paired ABBA/BAAB scheduling with thermal preflight settling and log-ratio confidence intervals produce PASS/FAIL/INCONCLUSIVE verdicts on sub-5% regressions.
 - Established first M5 MacBook Air community benchmarks, cited and adopted upstream by **RapidMLX** as official hardware reference data. Collaborated directly with upstream open-source maintainers.

SELECTED PROJECTS

- Thinking Budget — Adaptive Reasoning for LLMs** Python, PyTorch, GRPO, TRL, Qwen3 1.7B, OpenEnv
- Spun out of DeathClock's CVE dataset: designed an RL environment teaching a 1.7B model dynamic reasoning effort allocation across 150 real CVEs. Multi-objective reward for calibration, difficulty awareness, and adversarial robustness (defeated 5/5 attack strategies).
 - Full **SFT → GRPO** pipeline pushed calibration accuracy 33% → 88% and triage F1 0.14 → 1.00. **Global Finalist (Solo), Meta PyTorch OpenEnv Hackathon 2026 (70,000+ teams)**. Live on HuggingFace Spaces.

- DeathClock — Predictive CVE Risk Platform** Python, XGBoost, LightGBM, React, D3.js, VS Code Extension, Gemini
- Solo-built platform scoring codebases for undisclosed CVE likelihood ahead of public disclosure: gradient-boosted models trained on NVD CVE data, interactive React/D3.js risk dashboard, VS Code extension, and Gemini plain-English explanations. Drew external documentation request from an **Apple PM**.

- godmode — Zero-Config Agent Prompt Compiler & CLI** TypeScript, Node.js, sqlite-vec, BM25, Ink/React TUI, NPM
- Published CLI (@godmode/cli) auto-generating configs for 11 coding agents (Claude Code, Cursor, Copilot). Built local hybrid BM25 + sqlite-vec vector search compressing context windows from 1.2M to 87K tokens (**93% reduction**). Includes prompt compiler and Claude pre-hook interceptors.

- drift — Zero-Entitlement Apple Neural Engine Monitor** Python, XNU Kernel (ctypes), IOKit, Textual TUI, SQLite
- htop-style ANE monitor for Apple Silicon: reads per-PID `ri_neural_footprint` and `ri_energy_nj` via `proc_pid_rusage` without root. Detects active neural models via FD scanning. 140-test suite (~91% coverage), multi-threaded TOPS benchmark, AppleSMC thermals. Distributed on **Homebrew**.

- PARASITE EVOLVED — Codebase Security Organism** Tree-sitter AST, LangGraph, Genetic Algorithms, Three.js
- Multi-agent security tool combining Tree-sitter AST parsing across 5 languages, genetic attack mutation search, and automated patch synthesis, visualized in an interactive 3D graph. Scans 2,000+ Java files in ~4s. **Finalist (Solo), QuantCraft Hackathon**.

- ZenTorrent — Open-Source CLI Streaming Engine** Go, BitTorrent Wire Protocol, Homebrew, Winget, Chrome Extension
- Streams torrents directly into VLC by prioritizing initial 5% piece chunks via an embedded HTTP server. Multi-threaded peer-wire parsing and piece verification. **90+ GitHub stars, 50K+ Reddit views, #1 Trending on r/golang & r/commandline**. Distributed via Homebrew, Winget, and Chrome extension.

- VOID Zero — Local Agent Control Plane** TypeScript, Node.js, MLX, Apple Silicon
- \$0-runtime local control plane translating Anthropic Messages API to local MLX/OpenAI endpoints on Apple Silicon with asynchronous SSE streaming, memory-pressure gating, and proposal → critique → action deliberation loops.

ACHIEVEMENTS & HONORS

- Hackathon Honors (Solo): 1st Place Winner**, Trae AI Hackathon (#1 of all participants); **Global Finalist**, Meta PyTorch OpenEnv Hackathon 2026 (Thinking Budget, **70,000+ teams**); **Finalist**, QuantCraft Hackathon (PARASITE EVOLVED).
- Quantitative & Competitive Honors: IMC Prosperity Round 4** (delta-neutral options market maker, Black-Scholes, IV solver) & **Arbitrage Arena** (Pravega XII, IISc Bangalore).
- National Contests & Fellowships: Smart India Hackathon (SIH) Semi-Finalist** (ProjectPulse, MoSPI), **Adobe University Hackathon 2026** (R1 Coding/OA), **HackNova 2026 Rank 10 Finalist**, **IIT Madras AI/ML Excellence Program** (Microsoft Noida Campus cohort), **Google Solution Challenge 2026** (FairLens AI).
- Competitive Programming: LeetCode Rating 1808** (Top 8% globally), **Codeforces Pupil, AIR 18 DU Coding Contest**.

TECHNICAL SKILLS

- Languages:** C/C++, Go, Python, Rust, Swift, TypeScript, JavaScript, Java, SQL
- AI & Systems:** PyTorch, GRPO, RLHF, Hugging Face (TRL), LangGraph, Metal, MLX, llama.cpp, XNU Kernel, IOKit, Lock-Free Atomics
- Web & Tools:** React, D3.js, Three.js, Git, Docker, Linux/POSIX, Homebrew (tap author), Winget, FLINT/Arb, SymPy

EDUCATION & CERTIFICATIONS

- Galgotias University — B.Tech in Computer Science and Engineering (Core)** 2024 – Present
- Expected Graduation: 2028 | Greater Noida, India · **Certification:** DeepLearning.AI Specialization (Coursera / Andrew Ng)